

EnergyLens — Reality Gap & Data-Sourcing Plan

Carbon Accounting, EPC, CRREM, EU Regulation + the full data-honesty layer

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Purpose / Amaç

Companion to **EnergyLens_Study_Reference_EN.pdf** — specifically *Part II (Carbon Accounting, EPC, CRREM & EU Regulation)* and *Part V (the honesty layer)*.

For **every** number in EnergyLens that is currently an *assumption / indicative / proxy* value, this document gives three things:

1. **Now** — what the platform does today (the honest current state).
2. **Should be** — what the *real* standard or methodology requires, exactly.
3. **What to access** — the *real institution, dataset, API or agreement* we must connect to make the number production-grade — with the access route, the licence, and the cost.

All regulatory dates, factors and data-source facts below were **verified against primary sources in June 2026** (sources listed in §C). Where a value is a range or still a draft, it says so — nothing is invented.

tr — Bu doküman ne işe yarar: Worksheet’te “şu an indikatif / varsayım / proxy” diye işaretlediğim her sayı için → (1) şu an ne yapıyoruz, (2) gerçekte tam olarak nasıl olmalı, (3) bunu gerçeğe çevirmek için **hangi kurumdan / API’den veri çekeriz veya hangi anlaşmayı yaparız** (erişim yolu + lisans + maliyet). Hocaya da, freelance müşterisine de aynı tabloyu gösterebilirsin. Önce kısa tablo (§A), sonra her satırın detayı (§B), sonra kaynak rehberi (§C), en sonda “ne söylemeli” (§D).

§A — At-a-glance comparison table

The short table. Read top-to-bottom: the first 6 rows are **Part II** (carbon / EPC / CRREM / regulation); rows 7–11 are the rest of the **honesty layer** (data realism across the platform). “Effort” = how hard it is for us to close the gap, not how important it is.

#	Item	Now (what)	Should be (the real)	What we must access — who	Effort
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	m	we do)	standard)	route · cost	rt
1	Scope 3 emissions	(Scope1+Scope2) × 8% screening, scope3_disclosure_grade = False	GHG Protocol Scope 3 Standard — calculate the <i>material</i> categories from activity data × emission factors (buildings: Cat 13 downstream leased / tenant energy, Cat 1 embodied carbon, Cat 2 capital goods)	Tenant/sub-meter energy data (building owner , contractual) · emission-factor DB: DESNZ/DEFRA (free), ADEME Base Empreinte (free), ecoinvent (paid), EXIOBASE (free spend-based) · or Climatiq API (freemium)	High
2	Market-based Scope 2	Supplier-factor column exists but no contract data → equals location-based	GHG Protocol Scope 2 Guidance dual reporting: market-based from supplier-specific factor / Guarantees of Origin ; residual mix for untracked supply	Each building's electricity contract + GoO certificates (utility/supplier) · AIB European Residual Mix for the residual factor (free, annual)	Medium
3	Scope 1 (gas refrigerant)	Gas proxied from heating-share if no meter; refrigerant leakage excluded	Metered fuel × fuel factor; refrigerant fugitive = annual top-up (kg) × GWP (IPCC AR5/AR6). EU F-Gas Reg 2024/573 governs logs/leak checks	Gas meter / utility bills (owner) · refrigerant service logs / F-Gas logbook (HVAC contractor) · GWP values IPCC AR6 (free) · fuel factors DEFRA/DESNZ (free)	Medium
4	Grid electricity	DE (UBA 363 g/kWh, 2024) + TR (0.442) real & year-indexed; other countries IEA	Official annual location-based factor per country	EEA “GHG emission intensity of electricity” (free, per-country, EU) · IEA Emission Factors (paid, global) · ElectricityMaps (live/operational, freemium+commercial)	Low

	s (n on - DE /T R)	placeholder			
5	EP C ra ti ng s	Synthetic energy_certificate for B001–B010	Actual EPC from the building's Energieausweis / national register	UK EPC Open Data (free API/bulk) · DE : no central register yet → due ~May 2026 (EPBD Art 22); until then per-building from owner/issuer · EU Building Stock Observatory (aggregate only)	Medium
6	CR RE M pa th w ay s	Indicative 1.5 °C anchors in one swappable module	Official CRREM v2 pathways + emission factors , compared on Scope 1+2 basis	CRREM Library — open-access from ~1 Jul 2026 (free pathways + factors + Misalignment-Year guidance) · embedding into a commercial product sold to clients = CRREM License Partner agreement	Low (data) / Medium (commercial embed)
7	Re gu lat or y fa ct s/ da te s	“Current as of mid-2026”; Battery Reg mis-numbered 2023/1670 in app	Verified, sourced, and the app number corrected to 2023/1542	EUR-Lex (free) · EFRAG (ESRS) · national transposition trackers — see §B7 for the verified list	Low
8	U nd er lyi ng bu ild in g	Consumption/solar/battery synthetic for 10 buildings; weather/price/grid-CO ₂ real	Metered data from a real pilot building	A pilot (e.g. BSBI campus) · smart-meter feed (DE Smart Meter Gateway / utility portal) · BMS export (BACnet/Modbus) · utility-bill API	High

da				
ta				
9 H	Heating/	Sub-metered HVAC	Building sub-meters / BMS	Med
V	cooling/	circuits	points (facility) — a Tier-3 upsell	ium
AC	ventilation			
sp	modeled			
lit	(type coeff ×			
	HDD/CDD)			
1 Fo	MAPE is in-	Rolling-origin cross-	Nothing external —	Low
0 re	sample	validation (Prophet	code/method fix (more history	
ca		cross_validation)	helps)	
st				
ac				
cu				
ra				
cy				
1 Ba	Upper-	Price each discharged	ENTSO-E hourly day-ahead	Low
1 tt	bound (all	kWh by the hour it	prices (free — we already pull	–
er	discharge	displaces (hourly	this) — code/method fix	Med
y	valued at	dispatch vs hourly		
R	peak tariff)	price)		
OI				
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tr — tabloyu nasıl kullan: Hocaya tek sayfa olarak göster. Sol taraf “dürüst şu an”, orta “gerçek standart”, sağ “neye erişmemiz lazım”. En kritik 3 satır: **1 (Scope 3)**, **2 (market-based S2)**, **6 (CRREM)** — bunlar enerji/ESG akademisyeninin ilk soracağı yerler. Satır 7 (regülasyon) bir **düzeltilme** içeriyor: app’te **2023/1670** yazan yerleri **2023/1542** yap.

§B — Detailed reference (per item)

Each item: **the gap in one line, what “real” means, the exact source + how to connect, and the honest sentence to say.**

B1 — Scope 3 emissions

The gap. We report Scope 3 as a flat **8 % of (Scope 1 + Scope 2)** and explicitly flag it scope3_disclosure_grade = False. This is a *screening placeholder*, not a disclosure number.

What “real” means. The **GHG Protocol Corporate Value Chain (Scope 3) Standard** defines 15 categories. You do **not** report all 15 — you do a *materiality screen*, then calculate the material ones from **activity data × emission factor**. For a building portfolio the material categories are almost always:

- **Category 13 — Downstream leased assets** (the energy your *tenants* use in space you own and lease out). For most landlords this is the **single biggest** Scope 3 bucket.
- **Category 1 — Purchased goods & services** and **Category 2 — Capital goods** (the *embodied* carbon of construction materials and major refits).
- Plus Cat 5 (waste), Cat 6 (business travel), Cat 7 (commuting) where data exists.

Exact sources + how to connect:

- **Activity data** (the hard part): tenant electricity/heat consumption — obtained via **green-lease clauses**, tenant sub-metering, or the utility. This is a *data-access agreement*, not an API.
- **Emission factors** (the easy part):
 - **UK DESNZ / DEFRA “Greenhouse gas reporting: conversion factors”** — free, updated annually, the de-facto global default for activity-based factors.
 - **ADEME Base Empreinte / Base Carbone** (France) — free, official, large database.
 - **ecoinvent** — the gold-standard LCA database for embodied carbon (Cat 1/2) — **paid licence** (~€for a commercial seat).
 - **EXIOBASE** — free environmentally-extended input-output DB for **spend-based** estimates.
 - **Climatiq** — a commercial **API** that aggregates DEFRA/IEA/etc. — freemium, good for a productized integration.

Say this: “Scope 3 is currently an 8 % screening placeholder, clearly flagged as not disclosure-grade. To make it real I model the material categories — Category 13 tenant energy and Category 1 embodied carbon — from activity data times DEFRA/ADEME factors. The blocker is not the method, it’s getting tenant data via green-lease clauses.”

tr: Scope 3 = şu an %8 screening (etiketli, disclosure değil). Gerçeği: GHG Protocol Scope 3 Standardı, sadece *material* kategoriler (binalarda **Kat 13 kiracı enerjisi** + **Kat 1 gömülü karbon**). Faktör kolay (DEFRA/ADEME ücretsiz, ecoinvent paralı); zor olan kiracı verisi (yeşil kira sözleşmesi).

B2 — Market-based Scope 2

The gap. The engine has a `supplier_ef` column, but with no real contract data it returns the **location-based** number twice.

What “real” means. The **GHG Protocol Scope 2 Guidance** requires **dual reporting**: *location-based* (grid average) **and** *market-based* (your actual procurement). Market-based uses, in order of preference: a **supplier-specific** emission factor, an **Energy Attribute**

Certificate (in Europe a **Guarantee of Origin, GoO**), or — for the electricity you *didn't* cover with a certificate — the **residual mix**.

Exact sources + how to connect:

- **Supplier factor / GoOs** — from each building's **electricity supply contract** and the **Guarantee-of-Origin certificates** the utility issues. This is a per-building document, not an API.
- **AIB European Residual Mix** (Association of Issuing Bodies) — the **authoritative** residual-mix factor for European countries, **published free every year** at aib-net.org/facts/european-residual-mix, and explicitly aligned to the GHG Protocol market-based requirement.

Say this: *“The market-based mechanism is built; it just needs each building's supply contract and its Guarantees of Origin. For the uncovered portion I use the AIB residual mix, which is the European reference factor for exactly this purpose. Until I have the contracts, market-based equals location-based — and I say so.”*

tr: Market-based S2 mekanizması hazır; gerçek tedarikçi sözleşmesi + **GoO sertifikası** lazım, kalanı için **AIB Residual Mix** (ücretsiz, yıllık, Avrupa referansı). Sözleşme gelene kadar location-based'e eşit — bunu dürüstçe söyle.

B3 — Scope 1 (gas + refrigerant)

The gap. Where there's no gas meter we *proxy* Scope 1 from a heating-share of electricity, and we **exclude refrigerant leakage entirely**. Refrigerants (R-410A, R-32) have very high global-warming potential and are the classic *forgotten* Scope 1 source.

What “real” means.

- **Stationary combustion** (gas/oil boilers): metered fuel use × a fuel emission factor.
- **Fugitive / refrigerant emissions:** $\text{annual_refrigerant_topup_kg} \times \text{GWP}$. The top-up mass is the refrigerant a contractor adds to recharge a system — i.e. what *leaked*. **GWP** comes from **IPCC** (AR5 or AR6 100-year values; e.g. R-410A ≈ 2088). The **EU F-Gas Regulation (EU) 2024/573** (in force 11 Mar 2024) mandates leak checks and logbooks — so the data *exists* on site.

Exact sources + how to connect:

- **Gas/fuel data:** the building's **gas meter / utility bills** (owner).
- **Refrigerant data:** the **F-Gas logbook / maintenance records** held by the HVAC service contractor (mandatory under 2024/573).
- **GWP values:** **IPCC AR6** (free, authoritative).
- **Fuel factors:** **DEFRA/DESNZ** (free).

Say this: *“Scope 1 is solid where there's a gas meter. The honest gap is refrigerant fugitive emissions — I exclude them today rather than fake them. The data exists: the F-Gas logbook the contractor must keep under EU 2024/573, multiplied by the IPCC GWP. That's a clean next step.”*

tr: Scope 1 gaz sayacı varsa sağlam. Açık: **soğutucu gaz kaçağı** (şu an hariç — uydurmuyorum). Gerçeği: yıllık dolum (kg) × **IPCC GWP**; veri zaten **F-Gas logbook**'ta (EU 2024/573 zorunlu kılıyor).

B4 — Grid emission factors (countries beyond DE/TR)

The gap. Germany (UBA, year-indexed: 2022 = 433, 2023 = 386, **2024 = 363 g CO₂/kWh**) and Turkey (**0.442**) are real and sourced. Other countries are **IEA placeholders**.

What “real” means. A per-country, **annual, location-based** factor, ideally consumption-based (includes imports/T&D losses) for disclosure.

Exact sources + how to connect (pick by need):

- **EEA — “Greenhouse gas emission intensity of electricity generation” — free**, official, per-country for Europe (the EU average fell ~9 % in 2024 vs 2023). Best free option to replace placeholders across the EU.
- **IEA Emission Factors** — global, monthly/quarterly for 50+ countries — **paid subscription** (single/multi-user; the licence forbids third-party work — relevant if reselling).
- **ElectricityMaps** — **live / operational** grid intensity via API (freemium for non-commercial, commercial licence for production). Use for *operational* insight, **not** for the annual disclosure factor.

Keep the two apart: the **disclosure** factor stays annual-official (UBA/EEA/IEA). **Live** ElectricityMaps is for operational dashboards only — auditors want the fixed annual number.

tr: DE (UBA 363, 2024) + TR (0.442) gerçek; diğerleri placeholder. Ücretsiz çözüm = **EEA** (Avrupa, ülke-bazlı). Global için **IEA** (paralı). Canlı için **ElectricityMaps** — ama disclosure faktörü **yıllık-resmî** kalmalı.

B5 — EPC ratings

The gap. The energy_certificate (A+→G) for the 10 demo buildings is **synthetic**.

What “real” means. The building’s **actual EPC** (in Germany the *Energieausweis*, issued under GEG by a qualified issuer), pulled from a national register where one exists.

Exact sources + how to connect:

- **UK — EPC Open Data** (epc.opendatacommunities.org) — **free**, official, bulk download + API for England & Wales (Scotland separate). The cleanest open EPC dataset in Europe — good for a UK demo.
- **Germany — no central register yet.** Germany is the only large EU state without an EPC database; one is **due ~May 2026** under **EPBD Article 22**, which also requires free access to the full EPC for owners/tenants/managers. Until it’s live, EPC is a **per-building** document from the owner or issuer.

- **EU Building Stock Observatory** — free but **aggregate** (national stock statistics), not per-building — useful for benchmarks, not for a specific building's rating.
- Going forward, EPBD Art 22 forces every member state's EPC database to be accessible and to feed the BSO annually — so per-country EPC access will improve through 2026–2027.

tr: EPC şu an sentetik. Gerçeği: binanın **Energieausweis**'i. **UK** ücretsiz açık veri (demo için ideal). **Almanya'da merkezi kayıt yok** — ~Mayıs 2026 (EPBD Md. 22) geliyor; o zamana kadar bina-bazlı sahibinden. **EU BSO** sadece toplu istatistik.

B6 — CRREM pathways

The gap. Our decarbonization pathways are **indicative 1.5 °C anchors**, isolated in one module (PATHWAY_ANCHORS) so they can be swapped for official values.

What “real” means. The **official CRREM v2 pathways + emission factors**, with the comparison done on the correct scope (**CRREM uses Scope 1+2**, excludes Scope 3 — EnergyLens was already corrected to compare on this basis).

Exact sources + how to connect — note the 2026 change:

- The **free CRREM Excel tool retires 1 July 2026**. It is being replaced by an **open-access “CRREM Library”**: the **pathways, emission factors and Misalignment-Year computation guidance are published free** for open use. So the *raw data* to compute a correct stranding year becomes **freely available**.
- **But** embedding CRREM pathways/methodology **into a commercial platform sold to clients** still requires a **CRREM License Partner** agreement (crrem.org/license-partners). Nearly every major real-estate ESG platform is a License Partner.
- **Practical plan:** use the open CRREM Library values to replace our indicative anchors now (correct data, isolated module); take the **License Partner** step when EnergyLens is sold commercially.

Say this: “*My pathways are indicative today. CRREM is opening its dataset as a free library mid-2026, so I’ll swap in the official curves — they drop into one isolated module. For commercial resale I’d take the CRREM License Partner agreement, which is the standard route every ESG platform uses.*”

tr: Pathway’ler indikatif. **CRREM Library (1 Tem 2026, ücretsiz)** ile resmî eğrileri tek modüle takarım. **Ticari satışta License Partner** anlaşması (sektör standardı). Karşılaştırma **Scope 1+2**.

B7 — Regulatory facts (verified June 2026) + the one correction

This is what an academic will check. All verified against primary sources this month:

- **CSRD / Omnibus I** — the simplification directive was **published in the EU Official Journal on 26 February 2026** (Council adoption 24 Feb 2026; in force ~ 18 Mar 2026). It **narrows mandatory scope** to companies with **> 1,000 employees AND > €450 m net turnover**; the **listed-SME wave is removed**; the new scope applies for

financial years from 1 January 2027. Consequence for positioning: many mid-caps the old material called “Wave 2, filing FY2025 in 2026” are now **outside mandatory scope** — so position EnergyLens as “**ESRS-E1-aligned report-ready support**” and point smaller firms to the **voluntary VSME** standard. Do **not** say “everyone must comply.”

- **EPBD recast — Directive (EU) 2024/1275** — in force 28 May 2024; **national transposition due 29 May 2026. Non-residential MEPS:** member states set national standards by **1 January 2027**; worst-performing **16 % renovated by 2030, 26 % by 2033**; new buildings **zero-emission from 2030**.
- **EU Battery Regulation — Regulation (EU) 2023/1542** $\triangle$ **the app says “2023/1670” in places — fix it.** Industrial carbon-footprint declaration applies from **18 Feb 2026**; the **digital Battery Passport via QR from 18 Feb 2027** (industrial/EV/LMT > 2 kWh).
- **EU Taxonomy — Activity 7.7 “Acquisition and ownership of buildings.”** Original Climate Delegated Act: pre-2021 buildings qualify (climate-mitigation substantial contribution) with **EPC A or top-15 % of national stock** by operational primary energy demand; full alignment also needs **DNSH + minimum safeguards. 2026 draft revision** loosens this to **EPC D or top-50 %** — *draft, verify on adoption*. Never claim “Taxonomy-aligned” — only an **indicative alignment signal** for the SC criterion.
- **EU F-Gas — Regulation (EU) 2024/573** (in force 11 Mar 2024): leak checks, recovery, logbooks — the basis for refrigerant Scope 1 (B3).
- **EnEfG (Germany):** ISO 50001 / audit for ≥ 250 employees; energy-management system if annual final energy > **7.5 GWh. GEG §71:** heating installed from 2024 must be **≥ 65 % renewable** (a heat pump satisfies it); min U-values (wall ≤ 0.24 , roof ≤ 0.20 , window ≤ 1.30 W/m²K).

Access: all of the above are on **EUR-Lex** (free full text) and **EFrag** (ESRS standards, free).

tr: En kritik: **CSRD kapsamı daraldı** (>1000 çalışan & >€450M, FY2027) — “herkese zorunlu” DEME, “report-ready / VSME” de. **Battery Reg numarasını app’te 2023/1542 yap.** Taxonomy 7.7 2026 taslağı gevşetiyor (EPC D / top-50%, taslak). Hepsi **EUR-Lex**’te ücretsiz.

B8 — Underlying building data (synthetic → real)

The gap. Building-level **consumption / solar / battery** data is **synthetic** for B001–B010. The *external* signals are already real: **Open-Meteo/DWD** weather (211k real hourly rows feeding real HDD/CDD), **ENTSO-E** prices, **ElectricityMaps** grid CO₂.

What “real” means. Metered time-series from at least one **real building**.

Exact sources + how to connect:

- A **pilot building** — the cleanest path (e.g. the **BSBI campus**, which doubles as the EXIST/academic story).
- **Smart-meter data:** in Germany via the **Smart Meter Gateway** rollout / the utility’s customer portal; elsewhere via the DSO/utility API.

- **BMS export: BACnet/Modbus** points from the building automation system (the IoT adapters already exist).
- **Utility-bill ingestion:** CSV/API from the energy supplier.

Honest framing: synthetic *operational* data is normal and legitimate for a pre-customer demo — the *methods, factors and external signals are real*. State it plainly; an academic respects “synthetic operational layer, real methodology and real external data.”

tr: Bina tüketim/solar/batarya **sentetik** (henüz müşteri yok); hava/fiyat/grid-CO₂ **gerçek**. Gerçeğe geçiş = **pilot bina** (BSBI kampüsü), **akıllı sayaç** / **BMS export** / fatura API'si.

B9 — HVAC heating/cooling/ventilation split

The gap. The split is **modeled** from a building-type coefficient × the weather shape (HDD/CDD), so an office's HVAC share is essentially pinned by its type. A facilities engineer spots this instantly.

Should be: sub-metered HVAC circuits (or dedicated BMS energy points).

Access: building **sub-meters** / **BMS metering points** (facility hardware) — frame as a **Tier-3 upsell**: “*modeled split using the HDD/CDD method — sub-metering refines this.*”

tr: HVAC split modellenmiş (sub-metered değil). Gerçeği: **alt-sayaç** / **BMS noktası** (Tier-3 upsell).

B10 — Forecast accuracy metric

The gap. Forecast error (MAPE) is computed **in-sample** — testing the model on its own training data, which flatters accuracy.

Should be: rolling-origin cross-validation (Prophet's `cross_validation`) so the reported MAPE is out-of-sample and honest.

Access: none external — a **code/method fix**; more history per building improves it.

tr: MAPE in-sample → **rolling-origin cross-validation**. Dış veri gerekmez, kod düzeltmesi.

B11 — Battery ROI savings

The gap. Savings are an **upper bound** — all discharged energy is valued at the **peak tariff** and the demand saving is proportional to energy, not to peak-kW. So payback looks optimistic.

Should be: price each discharged kWh by **the hour it actually displaces** (hourly dispatch against the hourly price curve), and value demand-charge savings on the actual peak-kW reduction.

Access: ENTSO-E hourly day-ahead prices (free — already pulled) — a **method/code fix**, not new data.

tr: Batarya ROI **upper-bound** (hepsi peak tarifieden). Gerçeği: her deşarj kWh'ını **yerine geçtiği saatin fiyatıyla** değerle (ENTSO-E saatlik — zaten var). Kod düzeltmesi.

§C — Data-source directory (institution • what • access • cost)

Quick reference — the real institutions, grouped. Use this when the professor asks “*where would the data come from?*” or when scoping a freelance build.

Emission factors & carbon accounting

- **UK DESNZ / DEFRA — GHG conversion factors** — activity-based factors (fuel, refrigerant, travel). **Free**, annual. (*global default*)
- **ADEME — Base Empreinte / Base Carbone** (FR) — large official factor DB. **Free**.
- **ecoinvent** — LCA database for **embodied carbon** (Scope 3 Cat 1/2). **Paid licence**.
- **EXIOBASE** — environmentally-extended input-output (spend-based Scope 3). **Free**.
- **Climatiq** — API aggregating DEFRA/IEA/etc. for productized integration. **Freemium**.
- **IPCC (AR5/AR6)** — refrigerant & gas **GWP** values. **Free**.
- **GHG Protocol** — the Corporate, Scope 2 and Scope 3 **standards + calculation guidance**. **Free** (ghgprotocol.org).
- **AIB — European Residual Mix** — **market-based Scope 2** residual factor. **Free**, annual (aib-net.org).

Grid emission factors (electricity)

- **UBA (Umweltbundesamt)** — Germany annual grid factor (**363 g/kWh, 2024**). **Free**.
- **EEA** — per-country EU electricity carbon intensity. **Free**.
- **IEA — Emission Factors** — global, 50+ countries, monthly/quarterly. **Paid subscription** (resale-restricted).
- **ElectricityMaps** — **live** grid intensity API. **Freemium** (non-commercial) / **commercial licence** for production.
- **TEİAŞ** (TR) — Turkish grid operator; single canonical factor (~0.442). (*less granular than UBA*)

Prices & weather

- **ENTSO-E Transparency Platform** — EU day-ahead/hourly electricity prices + generation. **Free** API (email transparency@entsoe.eu, subject “Restful API access”).
- **DWD Open Data** — official German weather, **free incl. commercial** (legal mandate). *Use this for the commercial build.*
- **Open-Meteo** — free **non-commercial** only (CC BY 4.0, < 10k calls/day); **commercial = paid subscription or self-host** the open-source. (*we use it now; switch to DWD/self-host for sale*)

- **EPIAŞ / EXIST — Şeffaflık Platformu** (TR) — Turkish day-ahead market prices. **Free.**
- **MGM** (TR) — Turkish state meteorology. Weather for Turkish buildings.

EPC, buildings & decarbonization

- **UK EPC Open Data** (epc.opendatacommunities.org) — free EPC dataset (England & Wales) + API.
- **Germany central EPC register** — **due ~May 2026** (EPBD Art 22); until then per-building Energieausweis.
- **EU Building Stock Observatory** — free **aggregate** building-stock + EPC statistics.
- **CRREM Library** — open-access pathways + emission factors from **~1 Jul 2026** (free); **License Partner** for commercial embedding (crrem.org).

Regulation (primary text)

- **EUR-Lex** — full text of every directive/regulation (CSRD/Omnibus, EPBD 2024/1275, Battery 2023/1542, F-Gas 2024/573, Taxonomy). **Free.**
- **EFRAG** — the ESRS standards (incl. ESRS E1) + the voluntary **VSME**. **Free.**

Real building data (to replace synthetic)

- **Pilot building / BSBI campus** — metered consumption.
 - **Smart-meter feeds** — DE Smart Meter Gateway / utility portals; DSO APIs elsewhere.
 - **BMS export** — BACnet/Modbus (adapters already built).
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§D — What to say (professor + freelance clients)

The honesty framing (memorize the shape, not the words):

1. *“The methodology and the external data are real; the operational layer is synthetic until a pilot. Here’s exactly what I’d connect to make each number production-grade.”* — then point at §A.
2. *“Scope 1/2 are auditable today with row-level lineage from a sourced factor to the disclosed kg. Scope 3 is a flagged screening estimate, and market-based Scope 2 is built but waiting on supplier data — I label both honestly rather than overstate.”*
3. *“CRREM is indicative now; the official curves become a free open library in mid-2026 and drop into one module, with a License Partner agreement for commercial resale.”*
4. *“I keep the disclosure factor annual-official and use live grid data only for operational insight — because that’s what an auditor expects.”*

Why this wins with an academic: naming the exact gap and the exact source to close it is the strongest possible credibility signal. An academic rewards “I know precisely what isn’t real yet and precisely how to fix it” far more than a polished claim that everything is finished.

Freelance angle (same content, commercial spin): this table is a productized scope-of-work. Each “What to access” cell is a paid integration you can quote — “connect your AIB

residual mix + GoOs for audit-ready market-based Scope 2,” “wire your DEFRA factors for Scope 3 Cat 13,” “swap in licensed CRREM curves.” You sell *closing the gap*, building by building.

tr — özet: Hocaya güç veren şey “her şey bitti” demek değil, “**neyin gerçek olmadığını ve onu kapatmak için tam olarak hangi kaynağa bağlanacağımı biliyorum**” demek. Aynı tablo freelance’te **satış kapsamı:** her “neye erişmemiz lazım” hücresi ücretlendirebileceğin bir entegrasyon.

Verified June 2026 against EUR-Lex, EFRAG, UBA, EEA, IEA, AIB, CRREM Foundation, DWD, Open-Meteo, DESNZ/DEFRA, ADEME, ENTSO-E and the UK EPC register. Regulatory dates — re-check on EUR-Lex right before the meeting; the EU Taxonomy 7.7 loosening (EPC D / top-50 %) is a 2026 draft, not yet adopted.